



HG0024



Family-based association study between SNPs rs914458 and rs941798 (*PTPN1* gene) with type 2 diabetes mellitus

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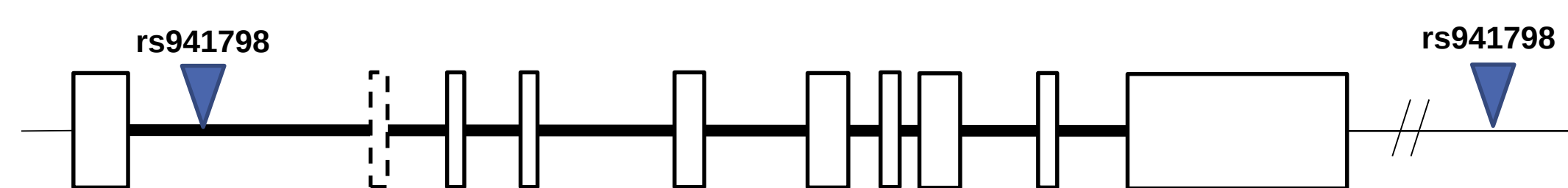
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Introduction

The prevalence of type 2 diabetes mellitus (T2DM) in Peru is 7%, while in Lima, the capital is up to 8.4%.¹

Although diet and lifestyle are important environmental factors in the development of T2DM, genetics factors have also been identified. Among them, *PTPN1* gene is one of the most widely studied genes due to its role in the insulin-signalling pathway.

Family designs, as opposed to population-based studies, avoid problems of population heterogeneity and stratification.²



The two SNPs studied here are located within the *PTPN1* gene locus on the chromosome 20 (20q13.13)

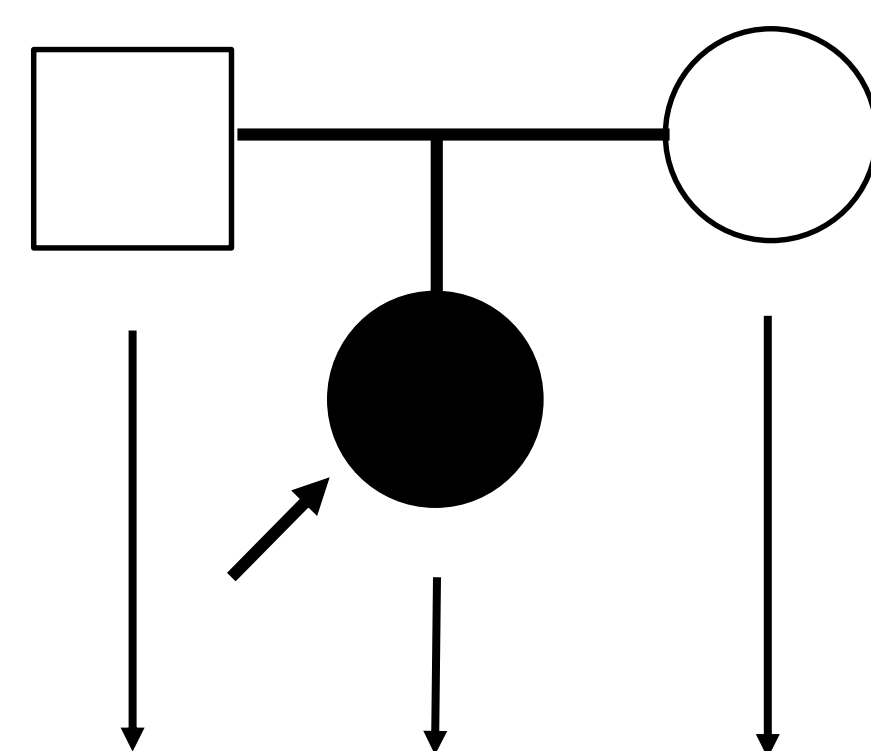
Objective

- The objective of this study was to investigate the association between SNPs rs914458 and rs941798 (*PTPN1* gene) with T2DM in a family-based association study in Lima, Peru.

Materials and Methods

Families description and DNA extraction and genotyping

- 41 family trios from Lima, Peru.
- Proband affected with T2DM.



- Blood samples

Genomic DNA extraction

- PCR and minisequencing
- SNPs: rs914458 and rs941798

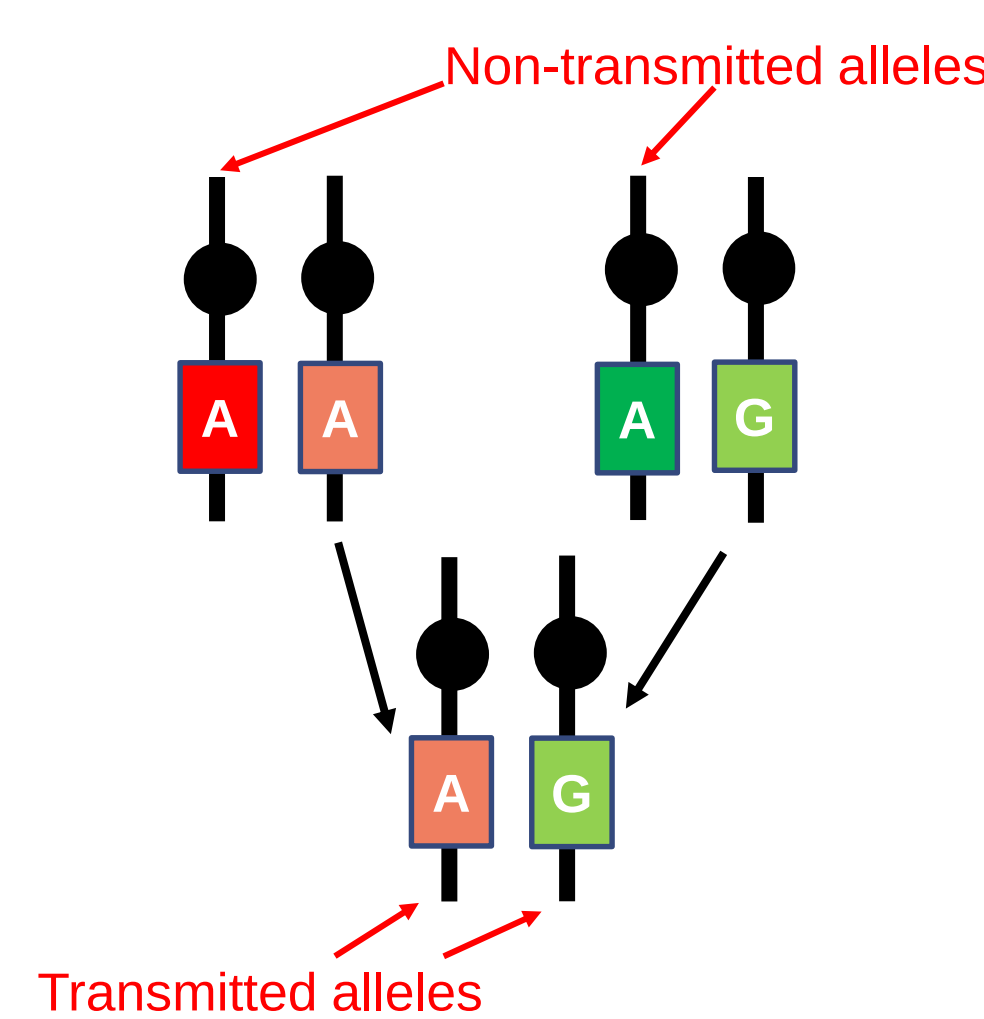
Amplification and genotyping

- SNPs characteristics

Haploview version 4.2

Association analysis

- Transmission-disequilibrium test (TDT)
- Transmitted alleles
- Non-transmitted alleles
- Is there differential transmission of alleles?



- If there is not linkage the proportion of transmitted alleles will be 0.5
- McNemar's test of marginal homogeneity
- H0: No linkage nor association between marker and disease

		Non-Transmitted alleles	
		A	G
Transmitted alleles	A	a	b
	G	c	d

$$\chi^2(1, N = n) = \frac{(|b - c| - 1)^2}{b + c}$$

Statistical softwares

R version 3.4.3

FBAT version 2.0.3

- SNP rs941798

		Non-Transmitted		Total
		G	A	
Transmitted	G	8	18	26
	A	22	4	26
		30	22	52

$$\text{McNemar's } \chi^2(1, N = 52) = 0.225, p = 0.635$$

Family-based association test (FBAT)

Marker	Bi-allelic mode		Model	Additive			Dominant			Recessive		
	Allele	Freq		N	Z	p	N	Z	p	N	Z	p
rs914458	G	0.81	14	1.414	0.16	5	1.5	0.13	13	0.866	0.39	
	C	0.19	14	-1.414	0.16	13	-0.866	0.39	5	-1.5	0.13	
rs941789	A	0.49	23	0.632	0.53	20	1.89	0.059	20	-0.882	0.38	
	G	0.51	23	-0.632	0.53	20	0.882	0.38	20	-1.89	0.059	

Neither the transmission disequilibrium test (TDT) nor the family-based association test (FBAT) under the additive, dominant or recessive model showed significant association of *PTPN1* gene SNPs with T2DM.

Results

SNPs characteristics

SNP	Alleles ^a	Sample	n	MAF ^b	HWE ^c
rs914458	G/C	Total	123	0.22	0.515
		Probands	41	0.17	1
		Parents	82	0.24	0.462
rs941789	G/A	Total	123	0.48	1.00E-04
		Probands	41	0.49	0.0885
		Parents	82	0.47	0.001

^a Major/minor allele

^b Minor allele frequency

^c Hardy-Weinberg equilibrium *p*-value

Transmission disequilibrium test (TDT)

- SNP rs914458

		Non-Transmitted		Total
		G	C	
Transmitted	G	32	11	43
	C	5	2	7
		37	13	50

$$\text{McNemar's } \chi^2(1, N = 50) = 1.563, p = 0.211$$

Discussion

Several studies show contradictory results regarding the association between several SNPs within the locus of *PTPN1* – including rs914458 and rs941789 – with T2DM and metabolic changes in different populations.

These conflicting findings may be attributed to the heterogeneity of T2DM aetiology given by differences in gene-gene and gene-environment interactions in the evaluated populations.

Although the major limitation in our study is the small number of families we believe that

There are limitations in our study such as the small number of families studied and the small number of SNPs evaluated. Thus we believe that further research taking into account other *PTPN1* gene SNPs in the Peruvian population is needed.

Second, our study included a small number of participants which may affect the reliability of results due to a lack of

Conclusions

-Based on our results we found no evidence of association between SNPs rs914458 and rs941798 (*PTPN1* gene) with T2DM in a family-based association study in Lima, Peru.

References

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